

KELLAR-STUDIO'S

AUTOMATION RUNBOOK & MAINTENANCE MANUAL

1. SYSTEM LIFECYCLE & PROCESS DAEMON INITIALIZATION

This runbook describes execution, continuous monitoring, and environment maintenance procedures for Kellar-Studio's background automation scripts. To prevent workflow disconnections, the `pipeline-watcher.js` instance must execute inside an isolated runtime container environment.

Process Initialization via PM2

To run the ingestion node script as an active background system service that boots automatically when your machine launches, use the standard system process manager:

```
# Install PM2 Process Daemon Infrastructure
npm install pm2 -g

# Boot Up the Kellar-Studio Ingestion Instance Engine
pm2 start pipeline-watcher.js --name "kellar-pipeline" --watch

# Enforce Automatic System Boot Persistence
pm2 startup
pm2 save
```

2. INFRASTRUCTURE LOGS & ENVIRONMENT MANAGEMENT

Monitoring engine behaviors and system calls requires analyzing centralized logging indices. Never alter environmental file strings directly inside operational folders while live render batches are printing.

- **Live Metrics Inspection:** Execute `pm2 logs kellar-pipeline` inside your target workspace console to view direct node file tracking events.
- **Security Separation:** Sensitive endpoints (such as publication webhooks or third-party storage access strings) must be stored inside a local `.env` file. The predefined `.gitignore` configuration blocks these files from remote visibility.

3. TECHNICAL TROUBLESHOOTING PROTOCOLS

When localized folder interruptions or memory allocation constraints block automated pipeline progression loops, run the following recovery operations sequentially:

Error Signature	Root Cause Vector	Resolution Mandate
File Lock Errors / EBUSY	The rendering timeline software is actively compiling frames, preventing the node script from capturing file arrays.	The script's built-in 3000ms stability parameter automatically retries. For high-density multi-gigabyte media arrays, scale the threshold limit to 10000ms.
Ignored File Output	The exported digital wrapper does not include the mandatory chronological syntax flags.	Re-verify file name structure against project templates. Append _FINAL prior to output drop.
Script Allocation Crash	Corrupted local dependencies or node runtime instance cache collisions.	Execute a structural flush and reboot instance: pm2 restart kellar-pipeline --update-env

4. SYSTEM MAINTENANCE ROUTINE

Maintain local system storage paths weekly to prevent unmanaged processing backlogs from consuming workstation memory blocks:

 **Regular Maintenance Schedule**

- Folder Purge:** Clear remnants from old processing operations inside 01_STAGE_FOR_RELEASE/ after confirming structural entries match your active text calendars.
- Dependency Re-validation:** Execute npm audit inside your repository framework quarterly to identify and update vulnerable node module arrays safely.